

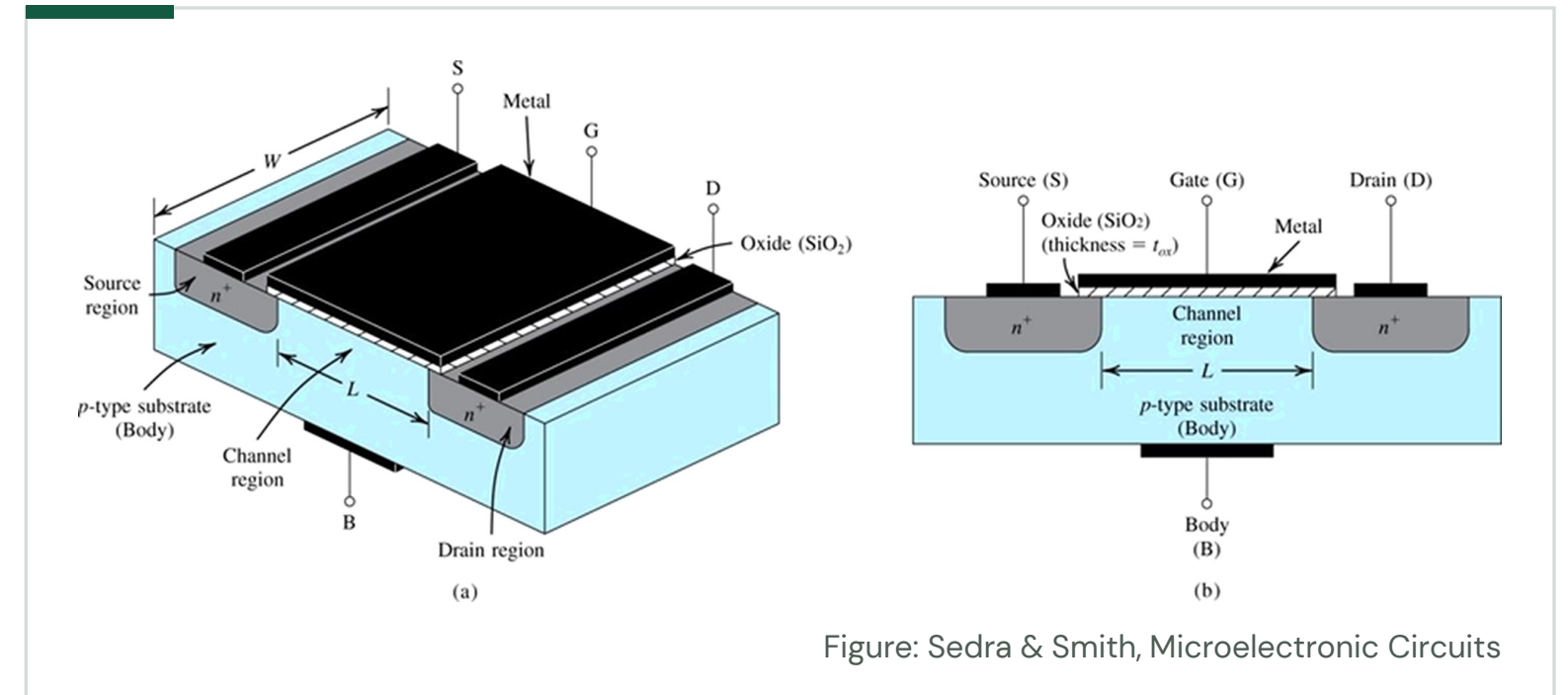
MOSFETS

Metal-oxide-semiconductor
field-effect transistors

LECTURE 3

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Computer Engineering · Academic Year 2025/2026 · Semester One



LECTURE MAP

1 **Structure and operation**
how a channel forms; the threshold voltage

2 **I–V characteristics**
cutoff, triode and saturation

3 **MOSFETs in DC circuits**
finding the operating point

4 **The MOSFET as an amplifier**
transconductance, small-signal model

5 **The MOSFET as a switch**

BY THE END YOU CAN

- Explain how v_{GS} creates a channel and how v_{DS} pinches it off
- Pick the right region and use its i_D equation
- Find the DC operating point of a MOSFET circuit
- Find g_m and the gain of a simple MOSFET amplifier

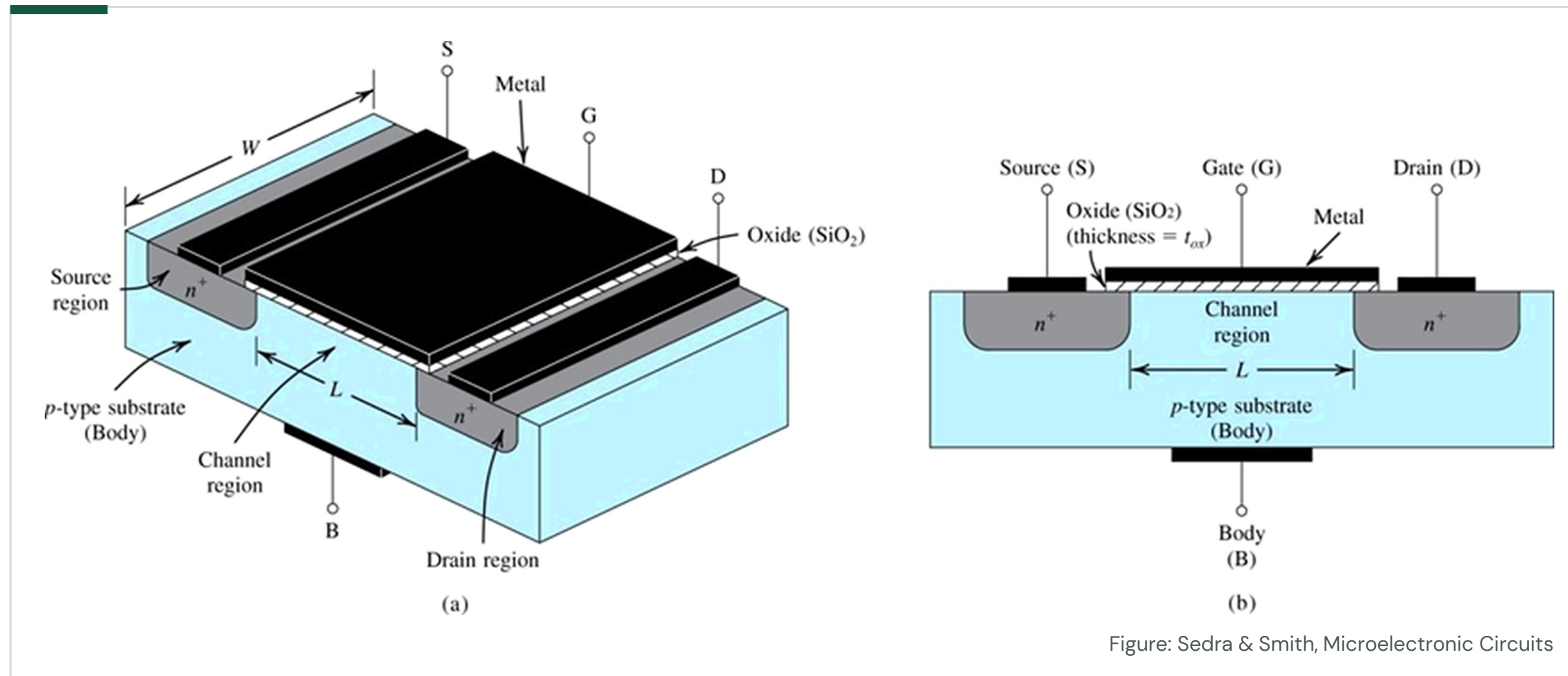
01

STRUCTURE AND OPERATION

- The device and its terminals
- No gate voltage: no current
- The threshold voltage
- Tapering and pinch-off
- CMOS; n- and p-channel symbols

- 1** Structure and operation
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THE N-CHANNEL ENHANCEMENT MOSFET



- Four terminals: **source** (S), **gate** (G), **drain** (D) and **body** (B).
- A thin oxide (SiO_2) insulates the gate, so the gate draws **no current**.
- The channel region between the n^+ source and drain has length L and width W .

WITH NO GATE VOLTAGE, NO CURRENT FLOWS

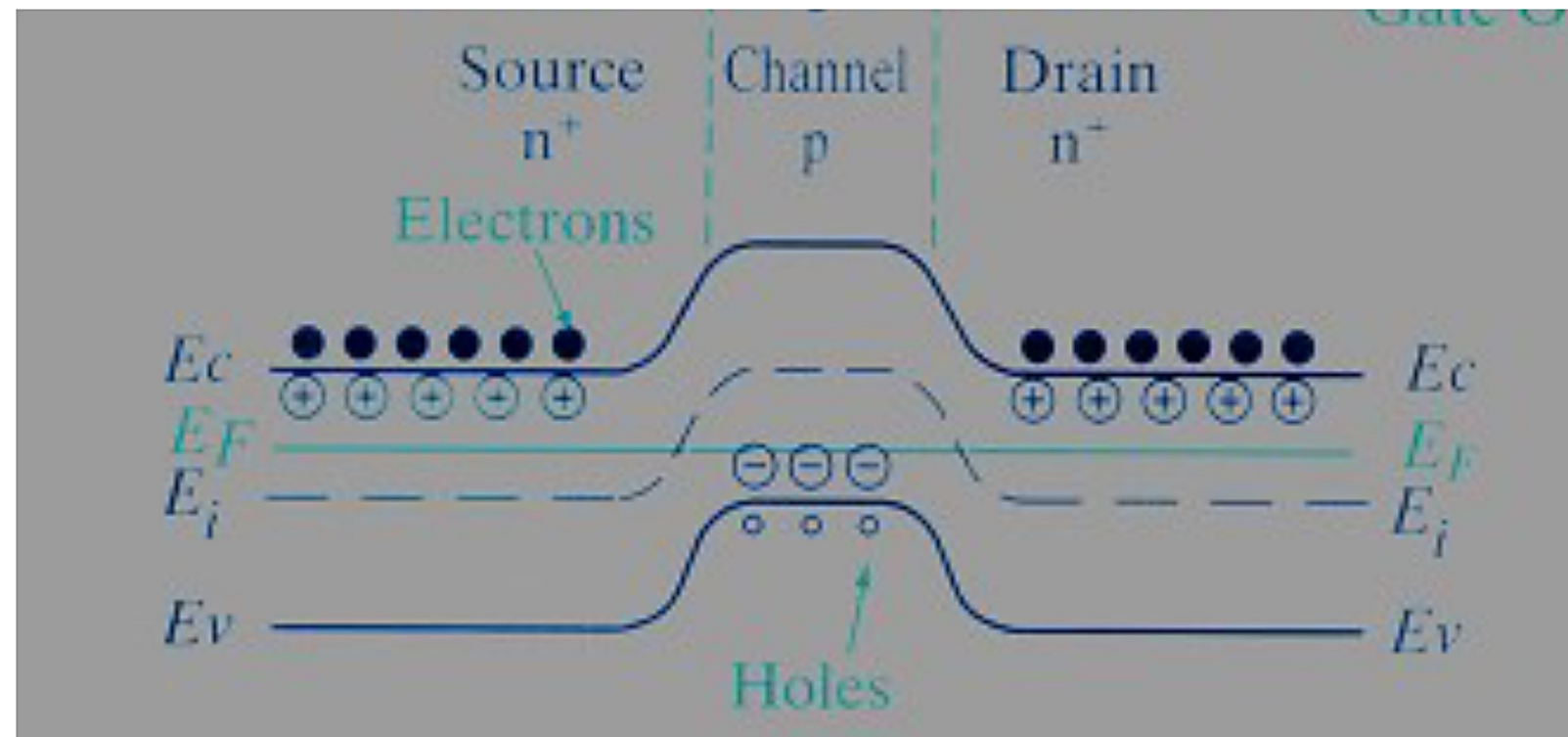


Figure: web source, via the original lecture slides

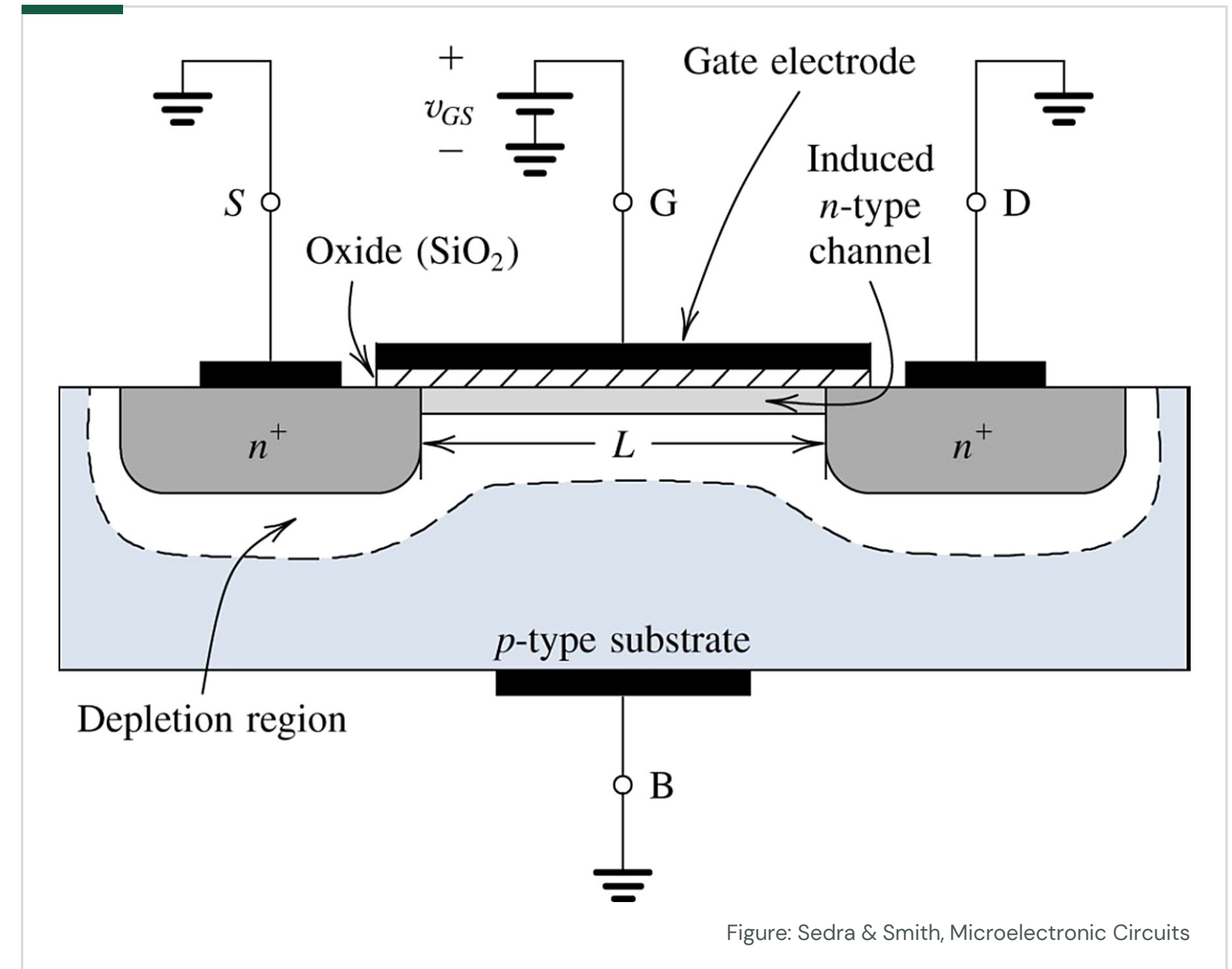
- Source and drain are n^+ , the body is p -type: two **back-to-back pn junctions** lie between source and drain.
- Whatever the polarity of v_{DS} , one junction is reverse-biased, so $i_D \approx 0$.
- The energy barrier (left) keeps the electrons of the source away from the drain.

CREATING A CHANNEL: THE THRESHOLD VOLTAGE

- A positive v_{GS} pushes holes away from the region under the gate and draws electrons in from the n^+ source and drain.
- Once enough electrons gather, they form an n -type **inversion layer**: the channel linking source to drain.

DEFINITION

Threshold voltage V_t : the value of v_{GS} at which the channel just forms.



INCREASING v_{DS} : THE CHANNEL TAPERS AND PINCHES OFF

- Along the channel, the voltage rises from 0 at the source to v_{DS} at the drain.
- The gate-to-channel voltage therefore falls from v_{GS} to $v_{GS} - v_{DS}$: the channel gets shallower toward the drain.

PINCH-OFF

When $v_{DS} = v_{GS} - V_t$, the channel depth at the drain end falls to almost zero. Increasing v_{DS} further barely changes i_D : the current **saturates**.

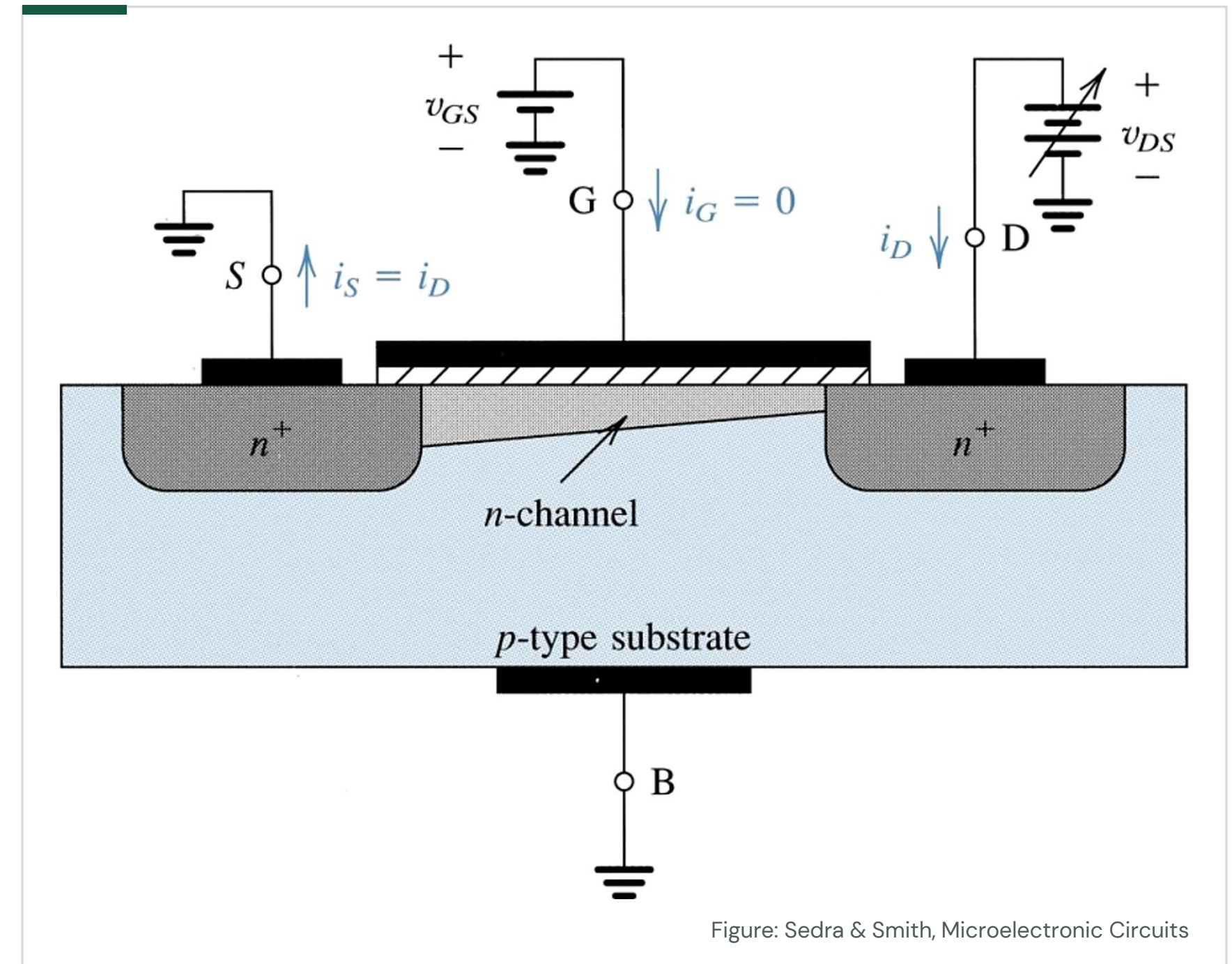
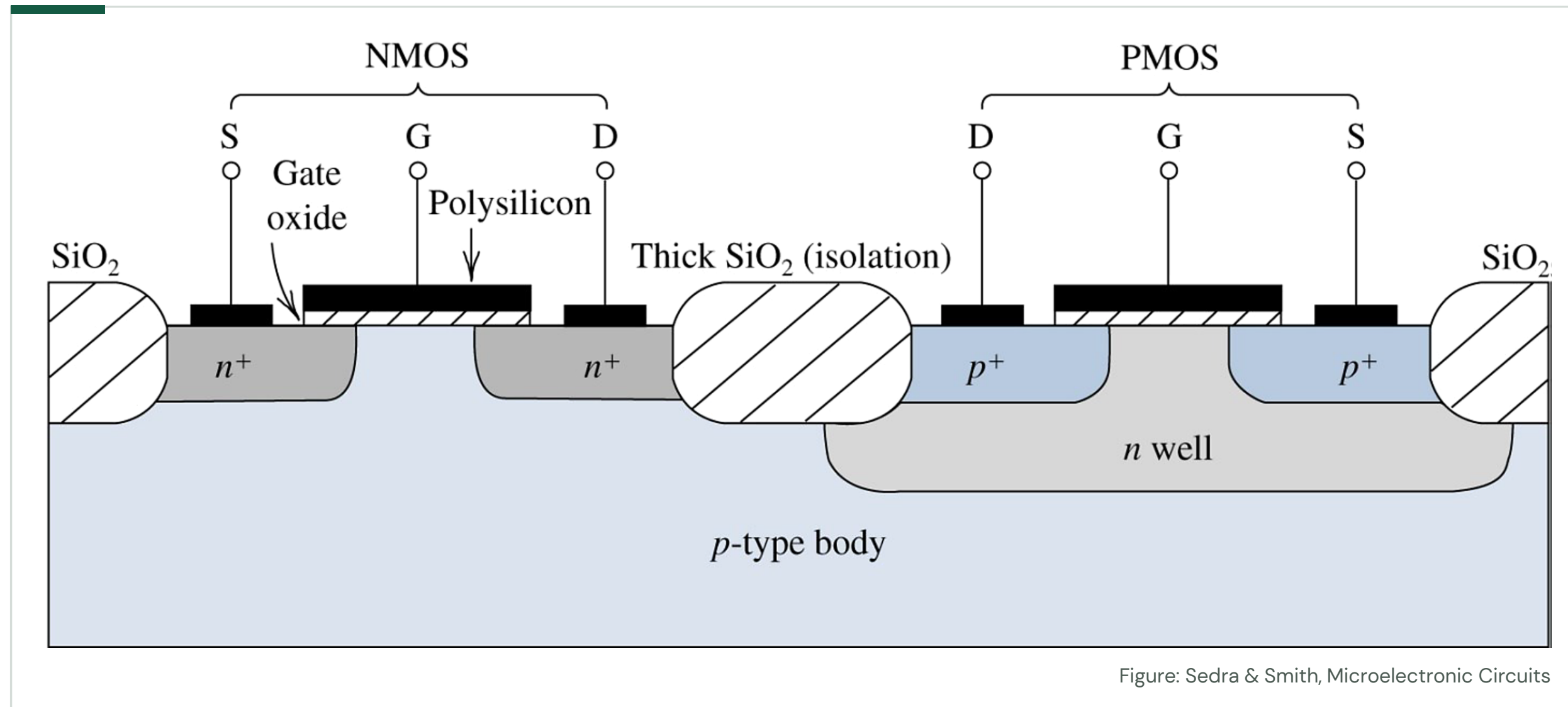


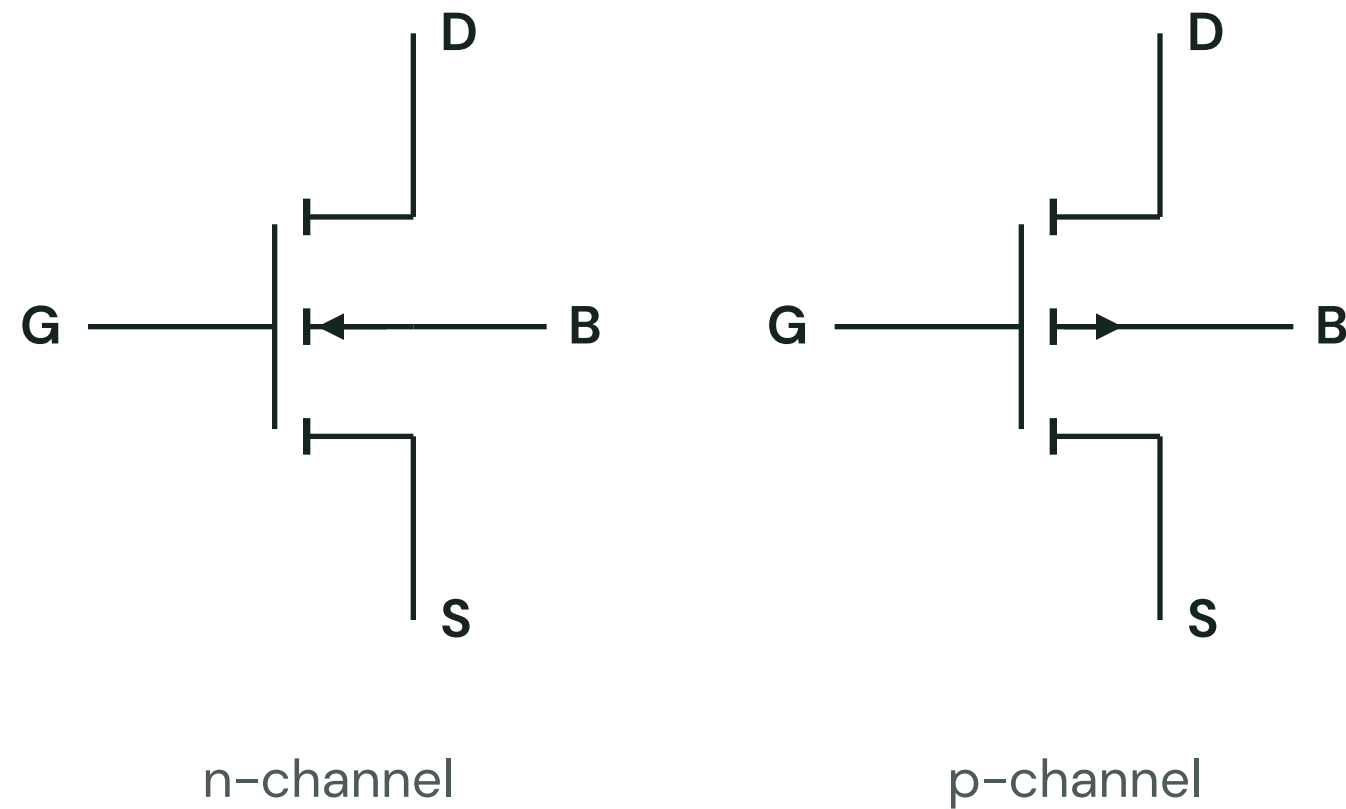
Figure: Sedra & Smith, Microelectronic Circuits

CMOS: NMOS AND PMOS ON ONE CHIP



- The PMOS transistor is built in an **n-well**, with p^+ source and drain.
- Thick oxide isolates neighbouring devices. CMOS circuits use both types together.

CIRCUIT SYMBOLS: N-CHANNEL AND P-CHANNEL



- In these symbols the arrow is on the body: it points **in** for an n-channel device and **out** for a p-channel device.
- PMOS: holes carry the current and every voltage and current is reversed. It conducts when $v_{GS} < V_t$, with V_t negative.

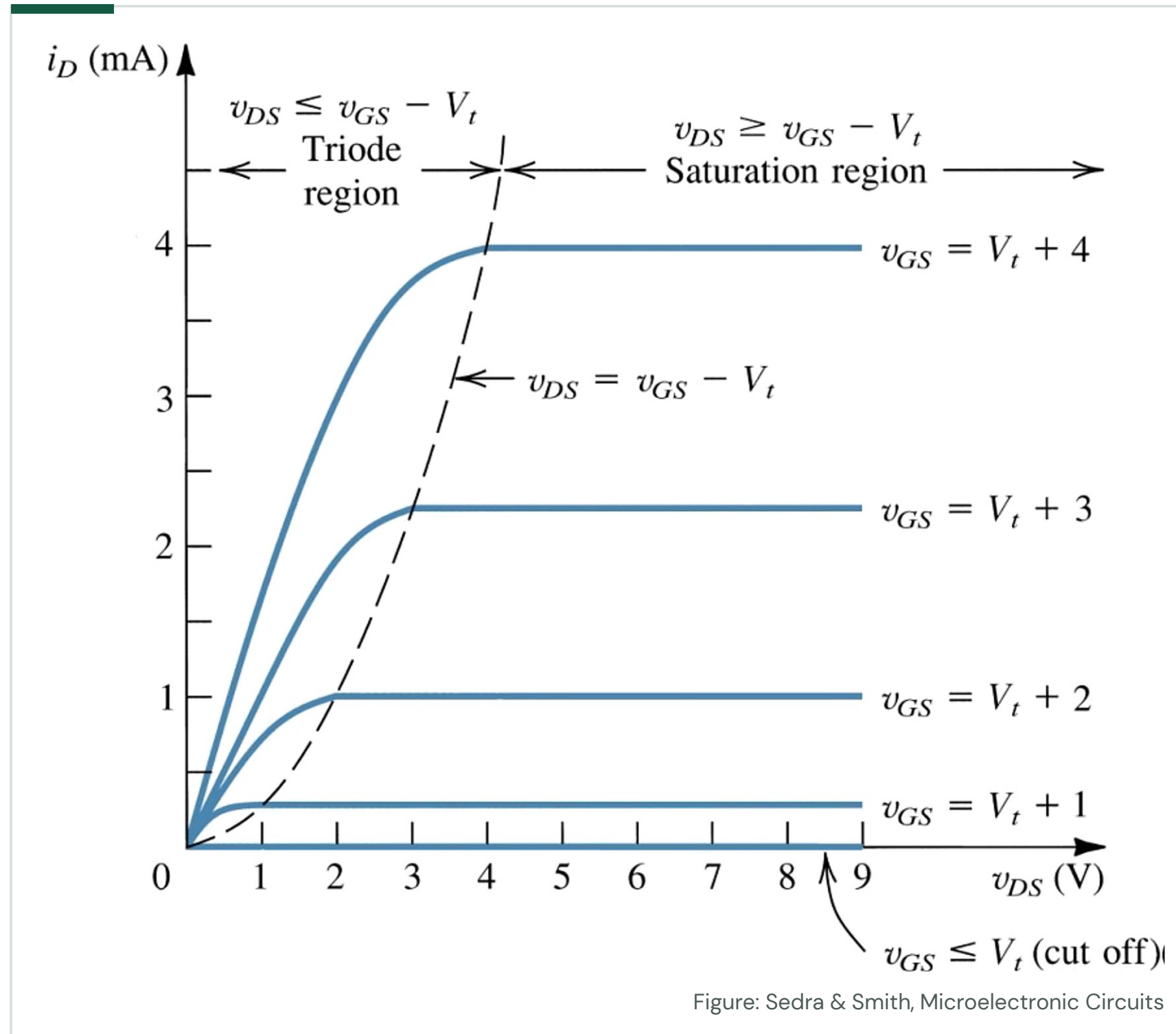
02

I-V CHARACTERISTICS

- The i_D-v_{DS} family
- Saturation: the square law
- Triode and saturation equations
- Resistor and large-signal models

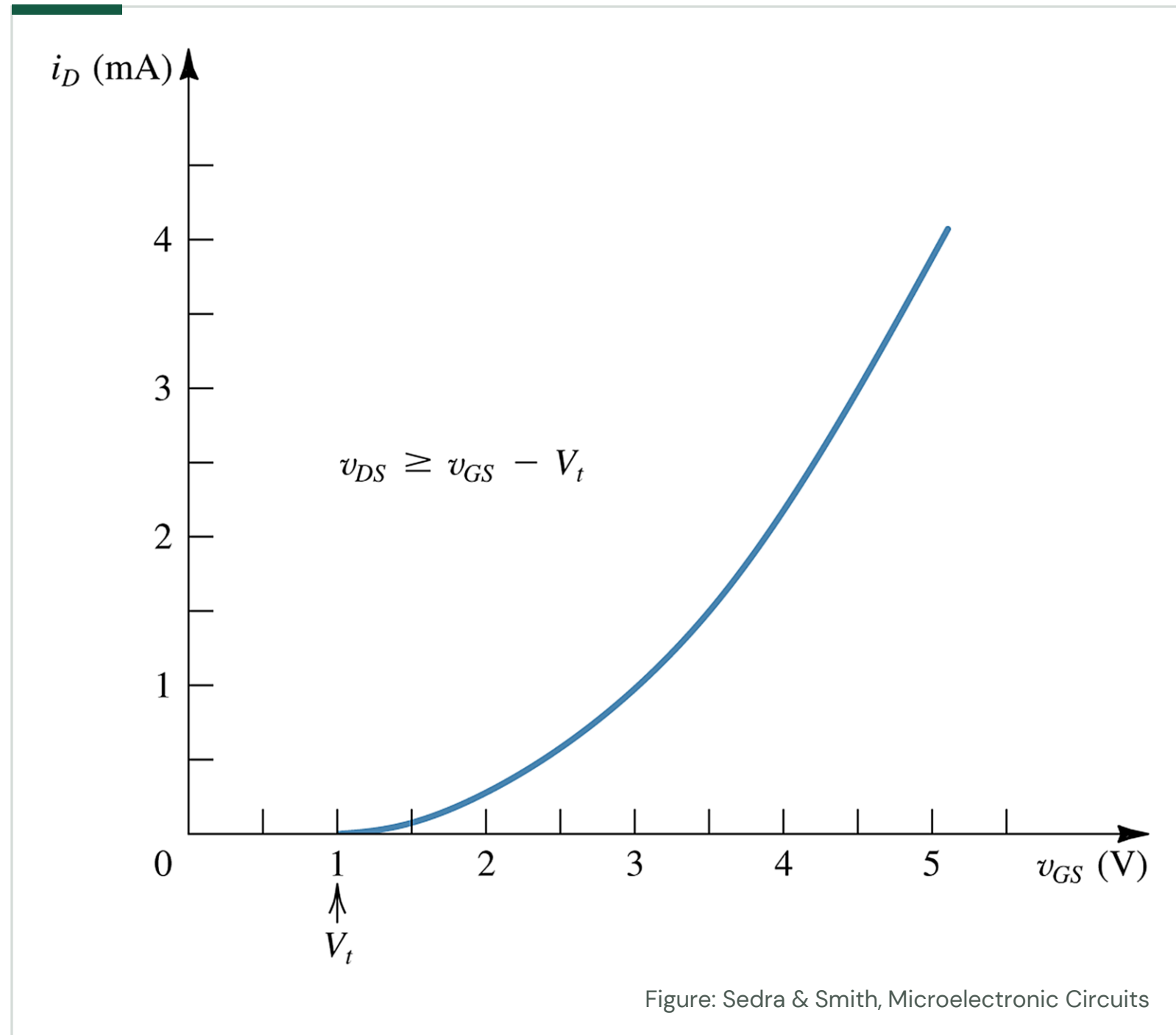
- 1 Structure and operation
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THE i_D - v_{DS} FAMILY OF CURVES



- Each curve is one value of $v_{GS} > V_t$.
- Left of the dashed line ($v_{DS} < v_{GS} - V_t$): **triode**. i_D rises with v_{DS} .
- Right of it: **saturation**. i_D is set by v_{GS} alone.
- For $v_{GS} \leq V_t$: **cutoff**, $i_D = 0$.

IN SATURATION, i_D DEPENDS ONLY ON v_{GS}

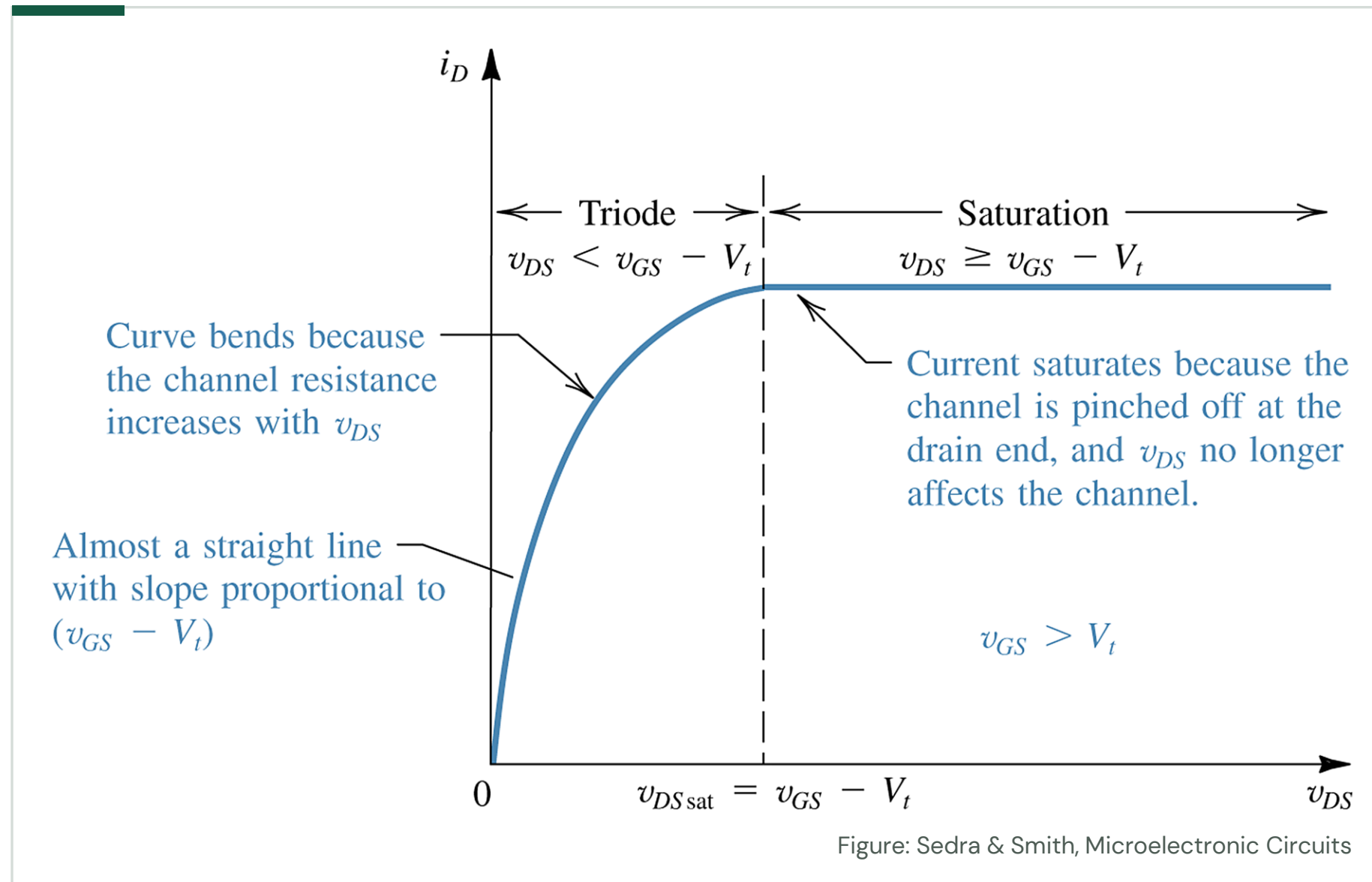


- Measured in saturation, the transfer characteristic is a parabola that starts at $v_{GS} = V_t$.

$$i_D = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2$$

- Doubling $(v_{GS} - V_t)$ quadruples i_D : the **square law**.

TRIODE AND SATURATION: ONE CURVE, TWO EQUATIONS



TRIODE $v_{DS} < v_{GS} - V_t$

$$i_D = k'_n \frac{W}{L} \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

SATURATION $v_{DS} \geq v_{GS} - V_t$

$$i_D = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2$$

Both require $v_{GS} > V_t$; without a channel, $i_D = 0$ (cutoff).

SMALL v_{DS} : A VOLTAGE-CONTROLLED RESISTOR

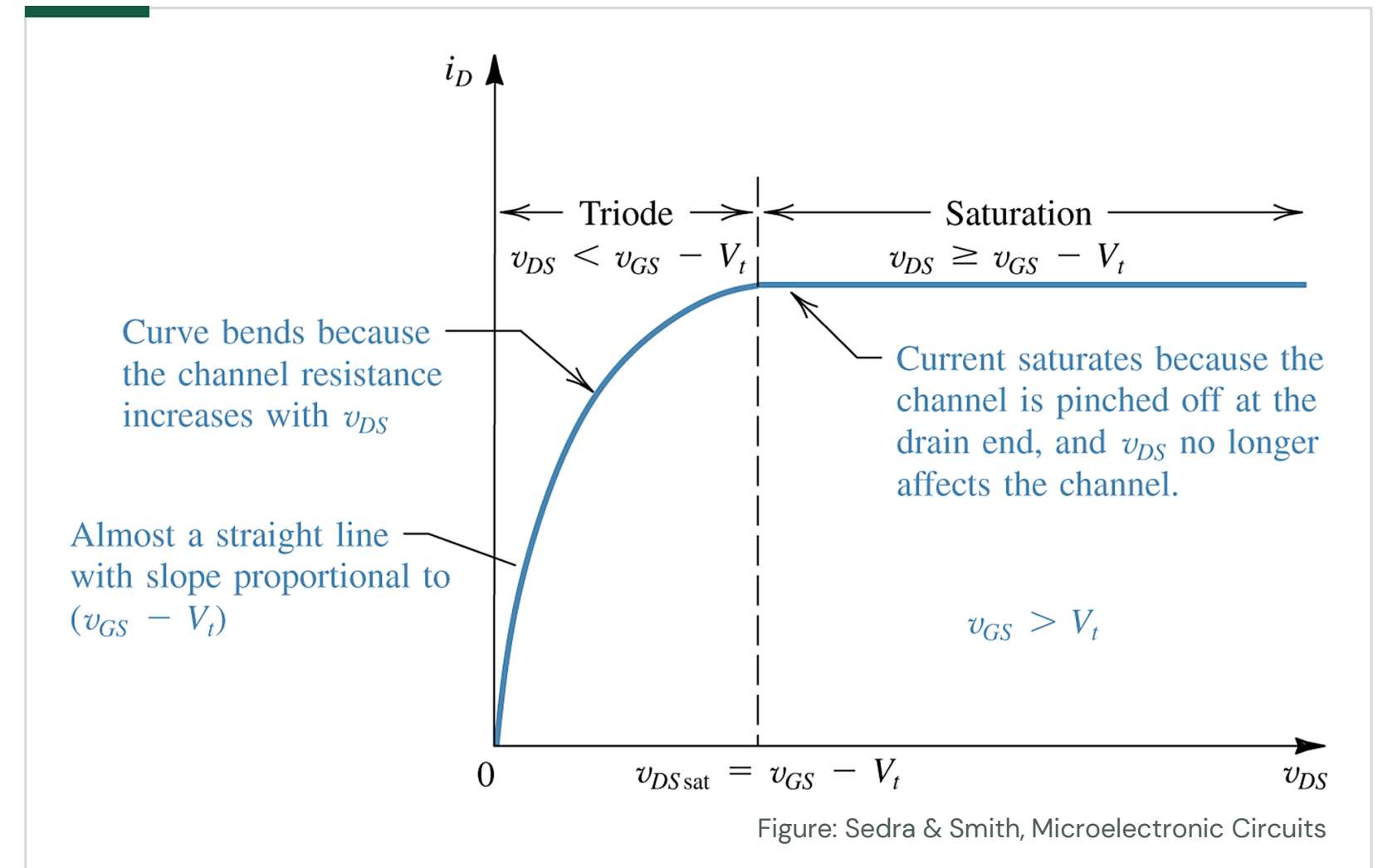
- For $v_{DS} \ll 2(v_{GS} - V_t)$ the v_{DS}^2 term is negligible:

$$i_D \approx k'_n \frac{W}{L} (v_{GS} - V_t) v_{DS}$$

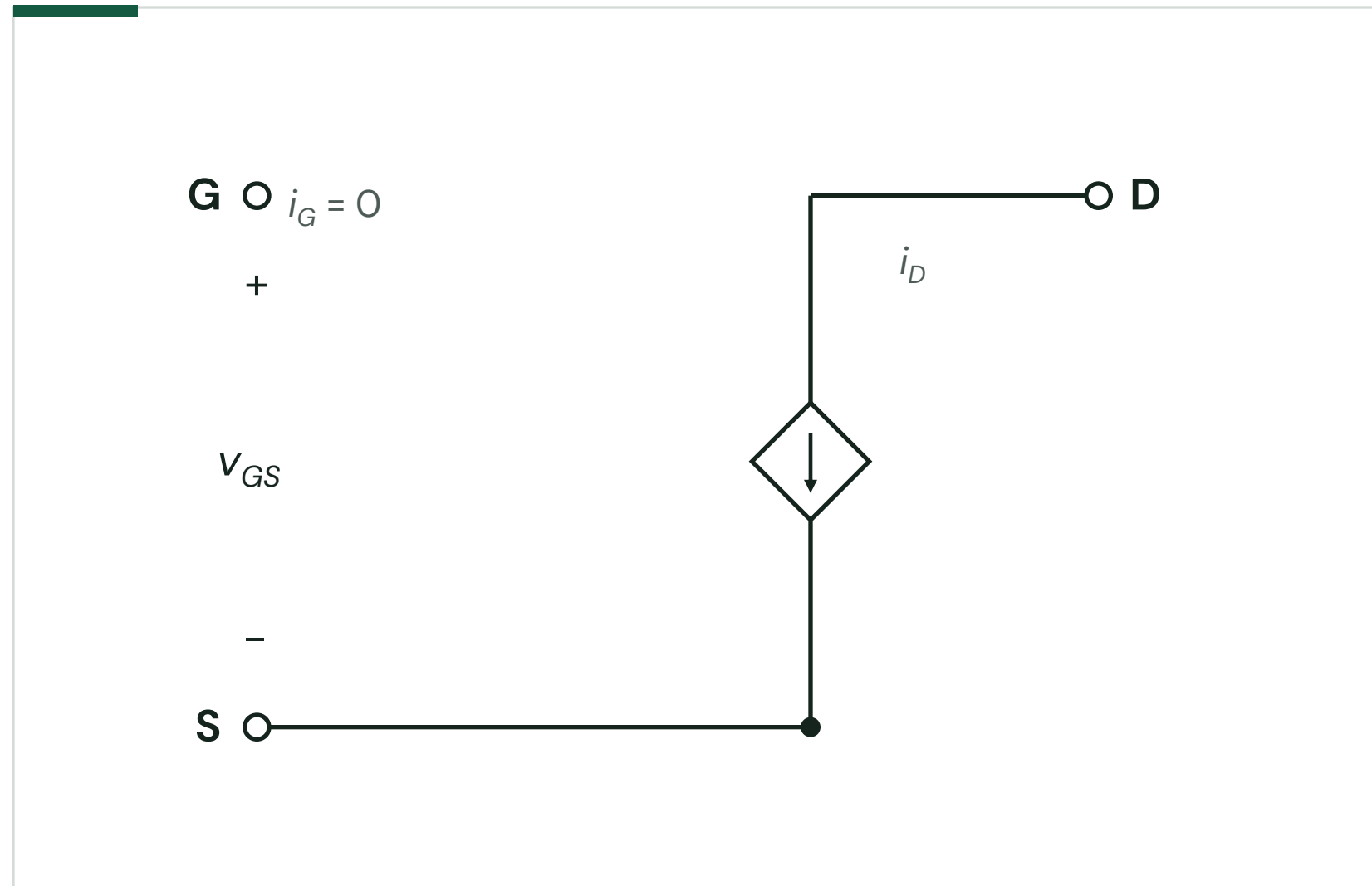
- The channel behaves as a resistor whose value v_{GS} sets:

$$r_{DS} = \left[k'_n \frac{W}{L} (v_{GS} - V_t) \right]^{-1}$$

- This is the region used when the MOSFET acts as a closed switch (Section 5).



LARGE-SIGNAL MODEL IN SATURATION



- In saturation the MOSFET acts as a **voltage-controlled current source** from drain to source.
- The gate draws no current: $i_G = 0$.
- The drain current is set by v_{GS} alone:

$$i_D = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2$$

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MOSFETS IN DC CIRCUITS

- Assume, solve, check
- Example: the drain voltage
- The current mirror

- 1 Structure and operation
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DC ANALYSIS: ASSUME, SOLVE, CHECK

1

ASSUME

a region, usually saturation (and $v_{GS} > V_t$).

2

SOLVE

with that region's i_D equation and KVL/KCL. Reject roots that give $v_{GS} < V_t$.

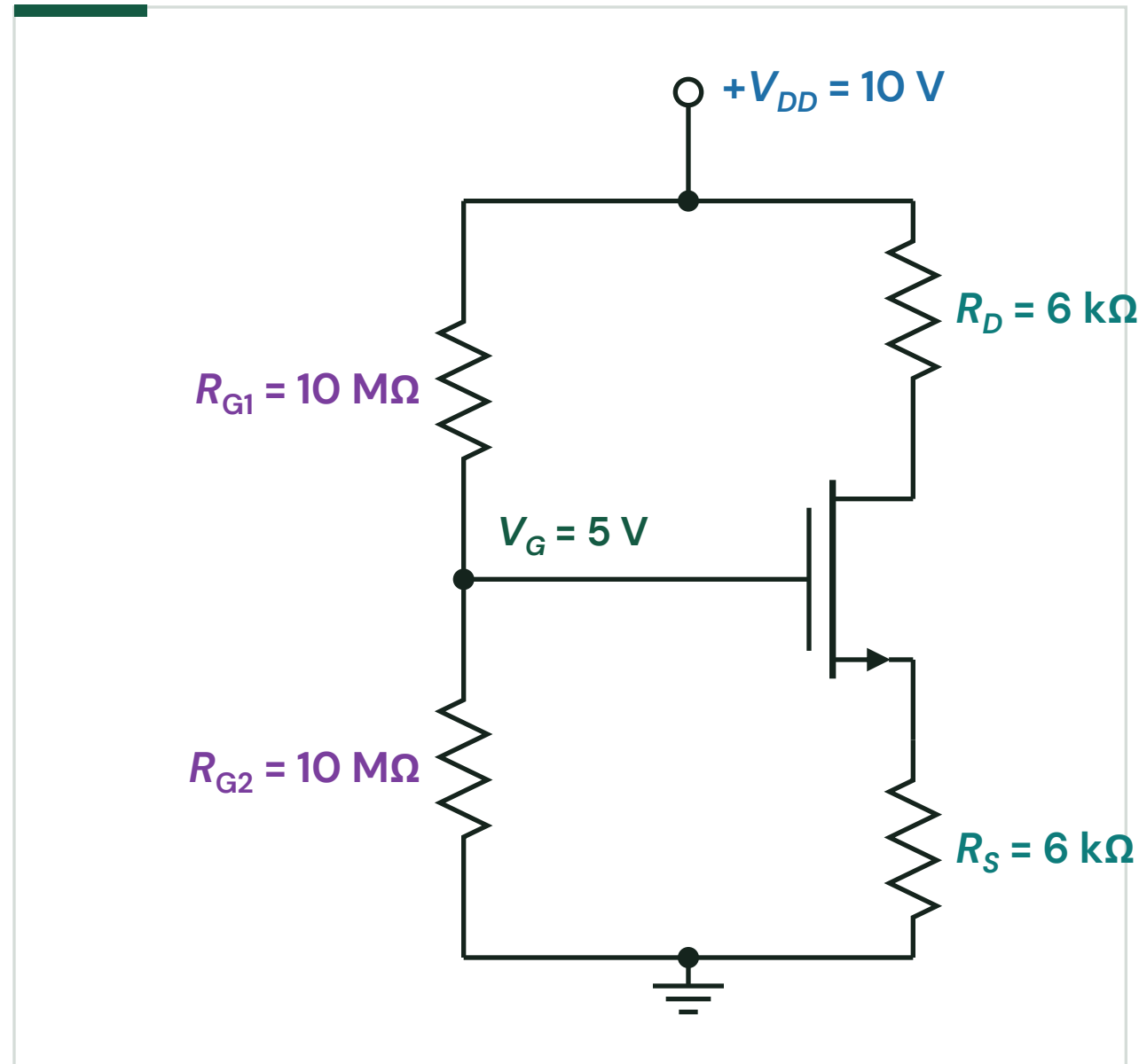
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CHECK

the assumption: saturation needs $v_{DS} \geq v_{GS} - V_t$. If it fails, redo it in triode.

The gate draws no current, so gate-bias resistors form an unloaded divider.

EXAMPLE 1: FIND THE DRAIN VOLTAGE (1 OF 2)



GIVEN

- $V_{DD} = 10 \text{ V}$
- $R_{G1} = R_{G2} = 10 \text{ M}\Omega$
- $R_D = R_S = 6 \text{ k}\Omega$
- $V_t = 1 \text{ V}, k'_n(W/L) = 1 \text{ mA/V}^2$

Find the drain voltage V_D . The MOSFET's region is not given.

1 Find the gate voltage V_G

No current flows into the gate, so R_{G1} and R_{G2} form an unloaded voltage divider.

$$V_G = V_{DD} \frac{R_{G2}}{R_{G1} + R_{G2}} = 10 \cdot \frac{10}{10 + 10} = 5 \text{ V}$$

2 Assume the MOSFET is in saturation

The region is unknown. Saturation is the usual bias region; we check this assumption at the end.

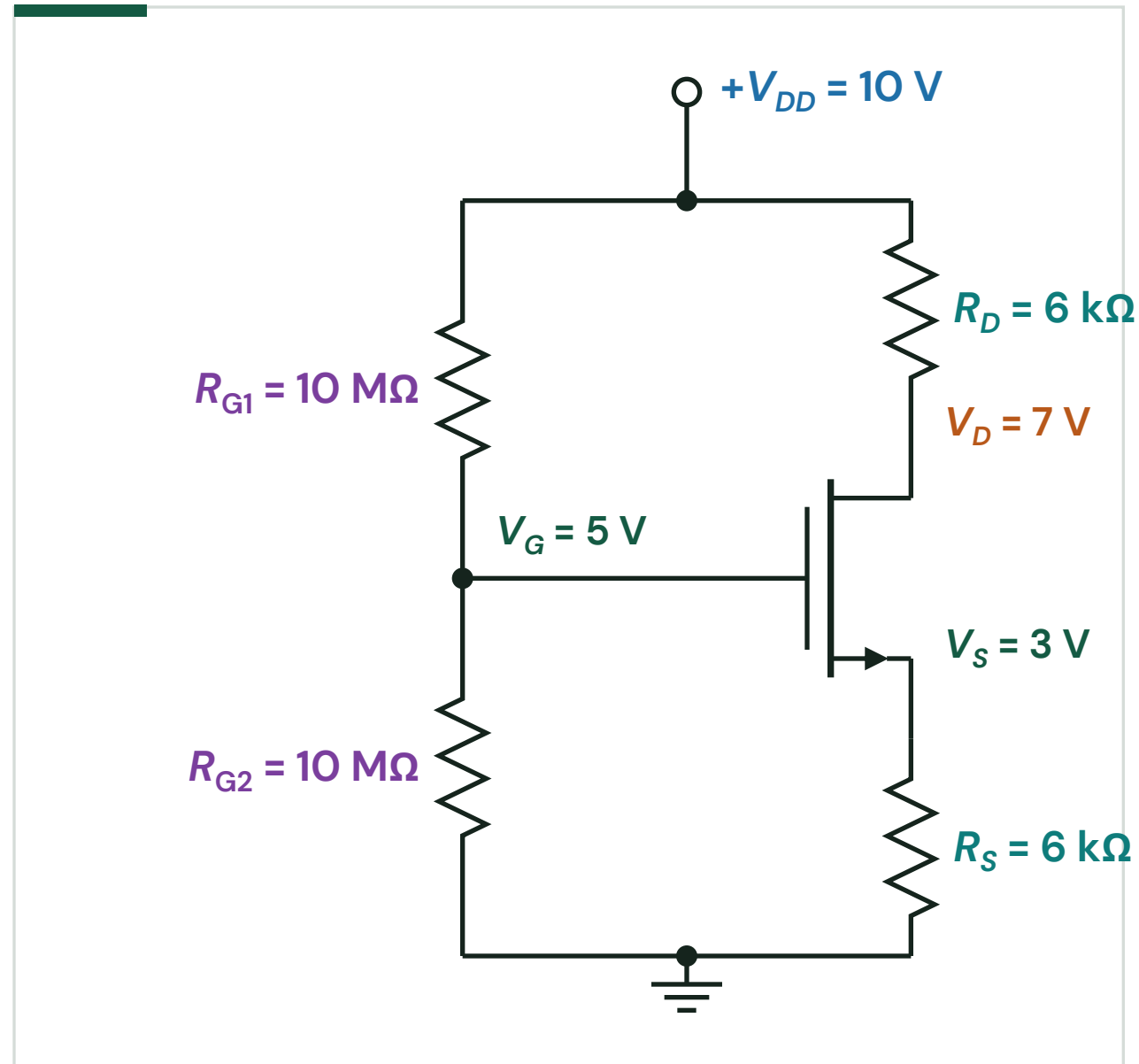
$$I_D = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_t)^2$$

3 Express V_{GS} through the source resistor

I_D flows through R_S lifting the source to $V_S = I_D R_S$. With I_D in mA and R in $\text{k}\Omega$, the products are in volts.

$$V_{GS} = V_G - I_D R_S = 5 - 6I_D$$

EXAMPLE 1: FIND THE DRAIN VOLTAGE (2 OF 2)



GIVEN

- $V_{DD} = 10\text{ V}$
- $R_{G1} = R_{G2} = 10\text{ M}\Omega$
- $R_D = R_S = 6\text{ k}\Omega$
- $V_t = 1\text{ V}$, $k'_n(W/L) = 1\text{ mA/V}^2$

4 Substitute V_{GS} into the saturation equation and solve

Units: I_D in mA. Expanding gives a quadratic with two roots.

$$I_D = \frac{1}{2} (1)(5 - 6I_D - 1)^2$$

$$18I_D^2 - 25I_D + 8 = 0 \Rightarrow I_D = 0.89 \text{ or } 0.5\text{ mA}$$

5 Keep the physical root

A valid solution must leave $V_{GS} > V_t$; otherwise there is no channel and no current.

$$I_D = 0.89: V_{GS} = 5 - 6(0.89) = -0.33\text{ V} < V_t \text{ (reject)}$$

$$I_D = 0.5: V_{GS} = 5 - 6(0.5) = 2\text{ V} > V_t \text{ (keep)}$$

6 Find the drain (and source) voltage

$$V_D = V_{DD} - I_D R_D = 10 - 0.5(6) = 7\text{ V}, \quad V_S = I_D R_S = 3\text{ V}$$

7 Check the saturation assumption

Saturation needs $V_{DS} \geq V_{GS} - V_t$

$$V_{DS} = 7 - 3 = 4\text{ V} \geq V_{GS} - V_t = 1\text{ V}$$

✓ The assumption holds.

$$V_D = 7\text{ V}$$

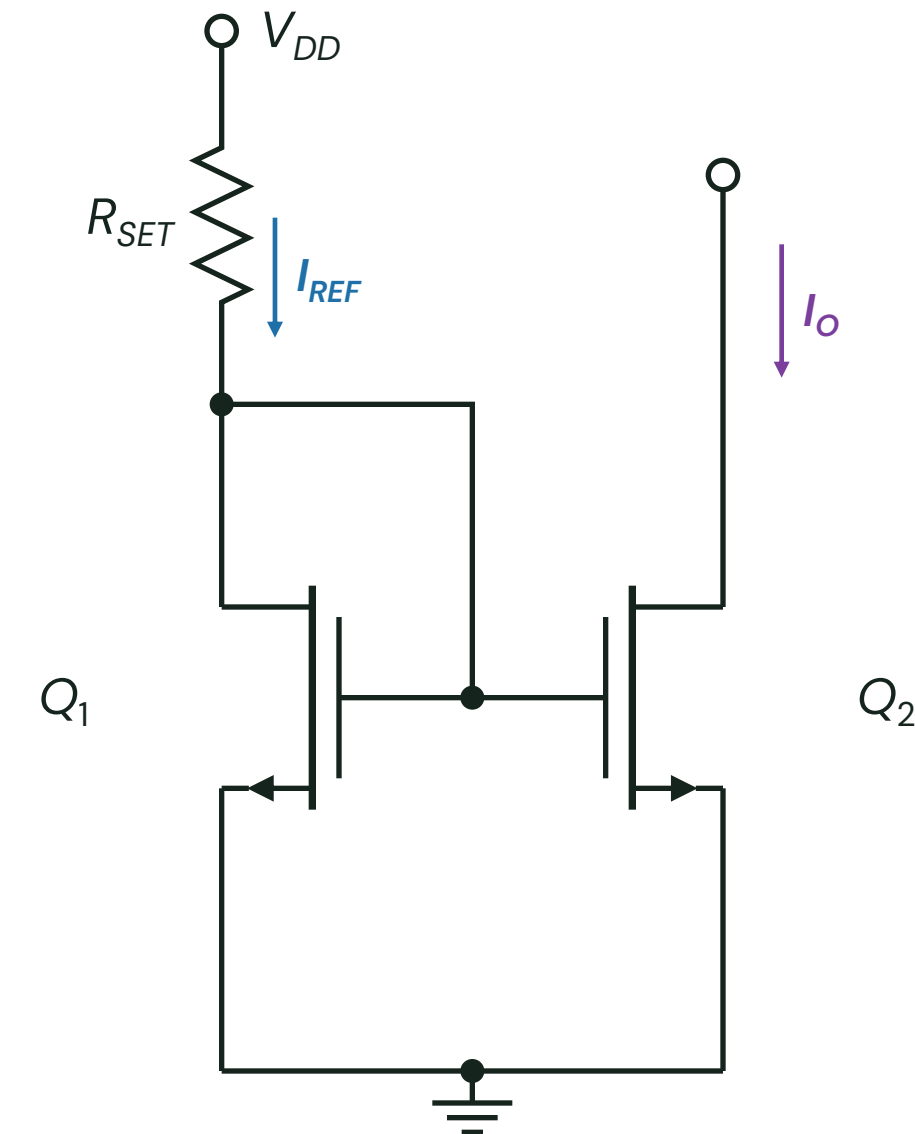
$$\text{with } I_D = 0.5\text{ mA}, V_{GS} = 2\text{ V}$$

THE CURRENT MIRROR

- Q_1 is diode-connected (gate tied to drain), so it is in saturation. R_{SET} sets its current I_{REF} , and with it V_{GS} .
- Q_2 has the same V_{GS} . If it is also in saturation, its current scales with W/L :

$$\frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$

- Identical transistors copy the current: $I_O = I_{REF}$.



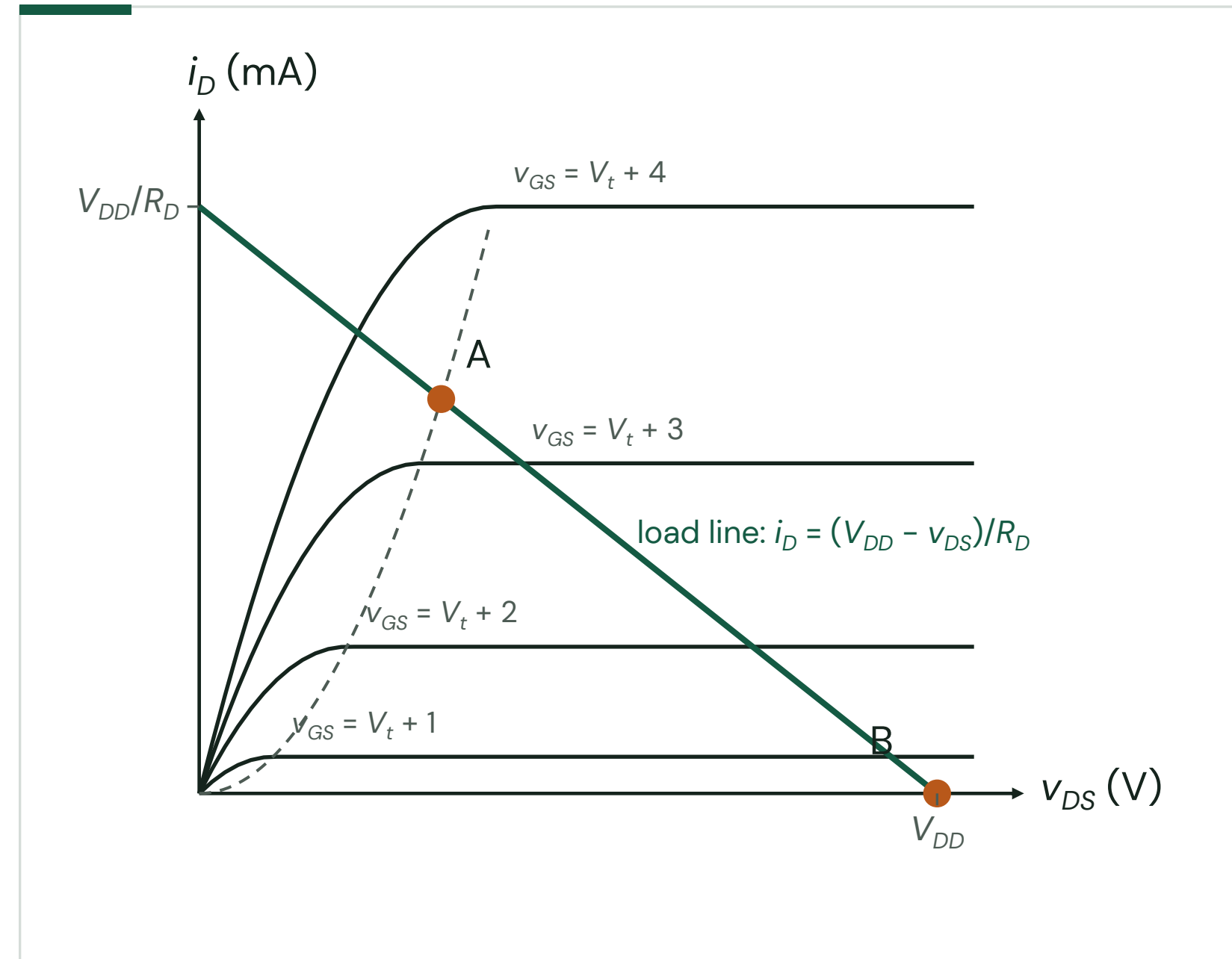
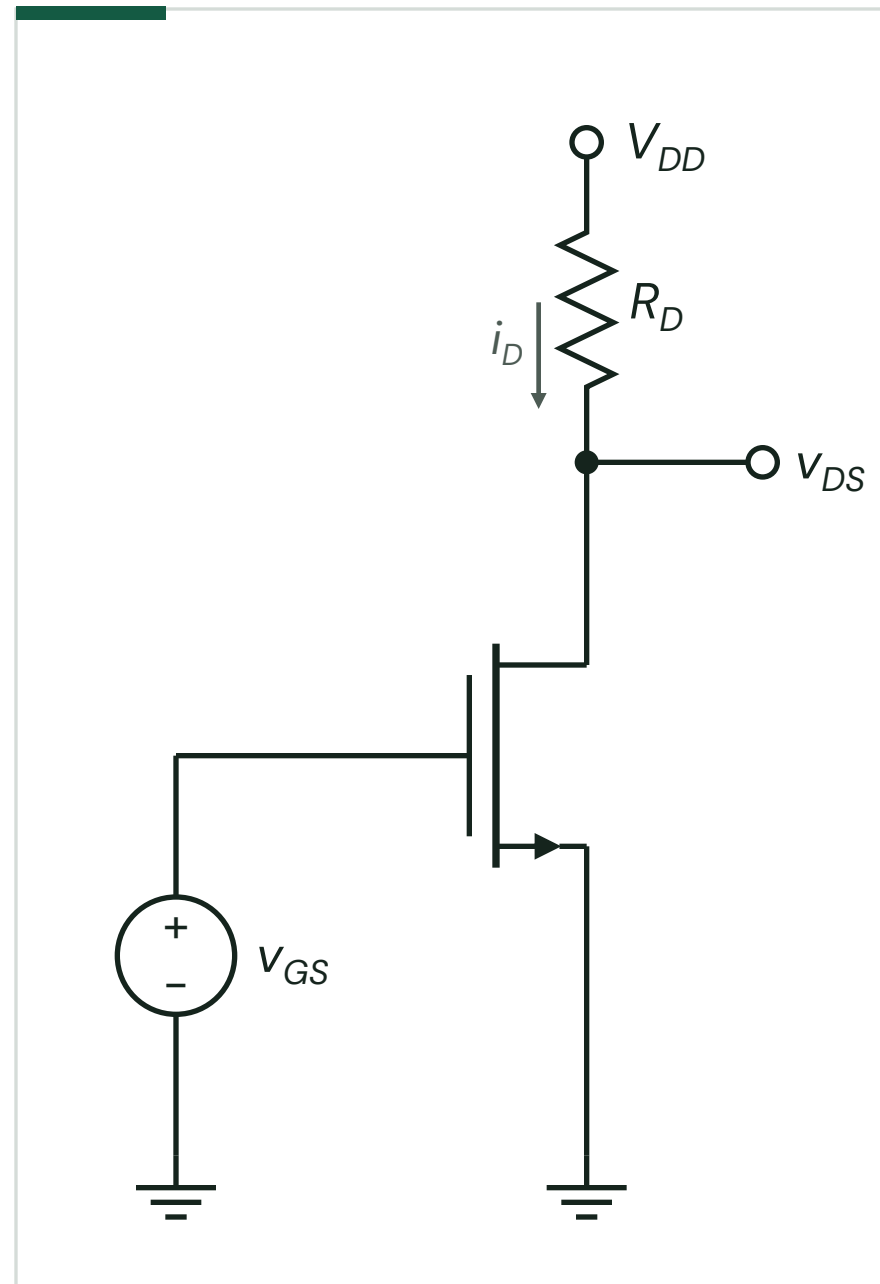
04

THE MOSFET AS AN AMPLIFIER

- The load line
- Transconductance g_m
- Small-signal models
- Example: gain and input resistance

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AMPLIFIER ACTION: THE LOAD LINE



Curves computed for $k'_n(W/L) = 1 \text{ mA/V}^2$, $V_{DD} = 10 \text{ V}$, $R_D = 1.25 \text{ k}\Omega$.

$$v_{DS} = V_{DD} - R_D i_D$$

- KVL at the drain: every operating point lies on this line.
- Between **A** (edge of triode) and **B** (cutoff) the MOSFET is in saturation: bias it there.
- A small change in v_{GS} moves the point along the line: a **large** change in v_{DS} .

SMALL-SIGNAL OPERATION: THE TRANSCONDUCTANCE g_m

- 1 The gate voltage is the bias plus a small signal

$$v_{GS} = V_{GS} + v_{gs}$$

- 2 Substitute into the saturation equation

$$i_D = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_t)^2 + k'_n \frac{W}{L} (V_{GS} - V_t) v_{gs} + \frac{1}{2} k'_n \frac{W}{L} v_{gs}^2$$

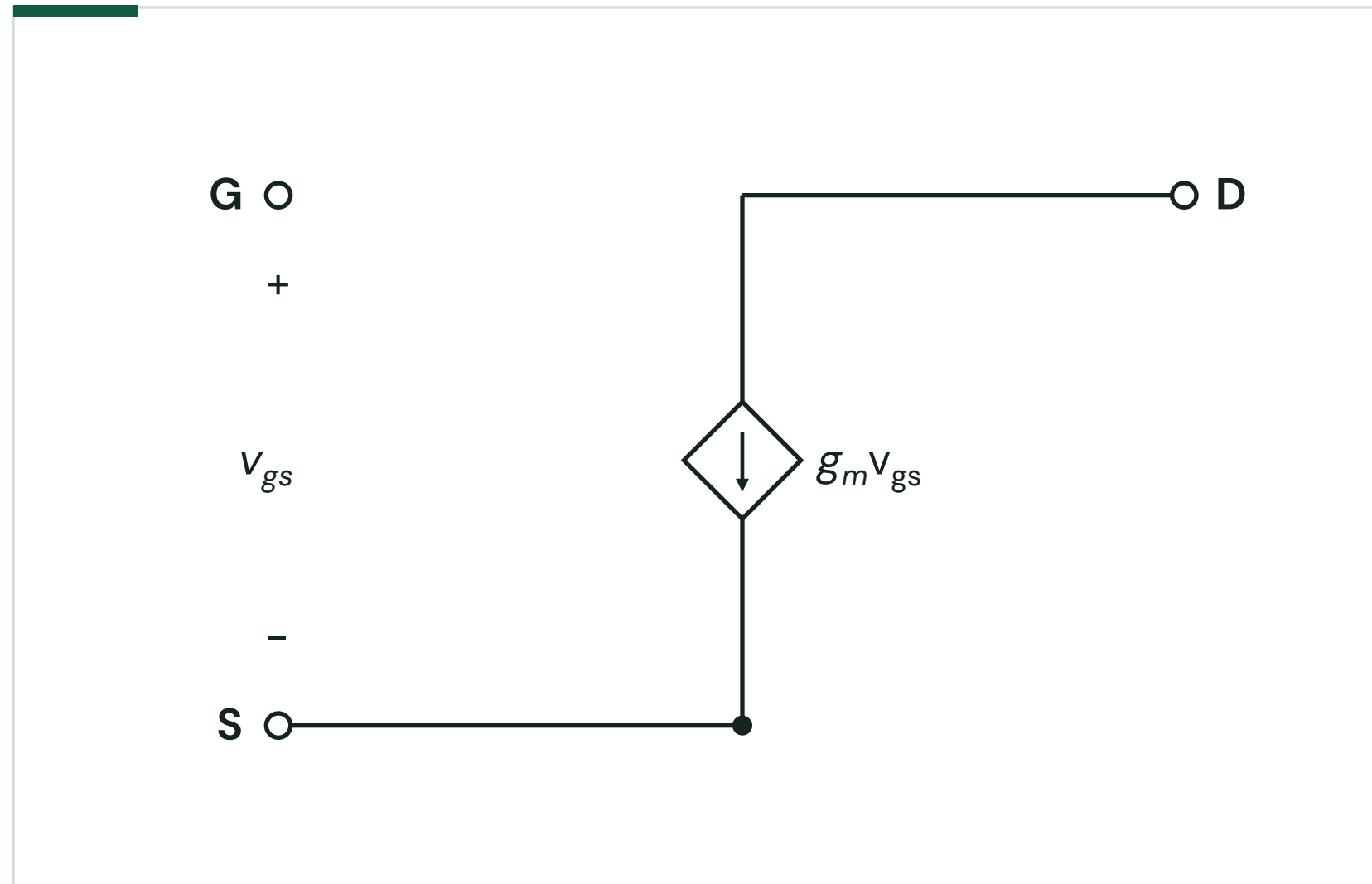
- 3 If $v_{gs} \ll 2(V_{GS} - V_t)$, the last term is negligible: the signal current is linear in v_{gs}

$$i_d = g_m v_{gs}, \quad g_m = k'_n \frac{W}{L} (V_{GS} - V_t)$$

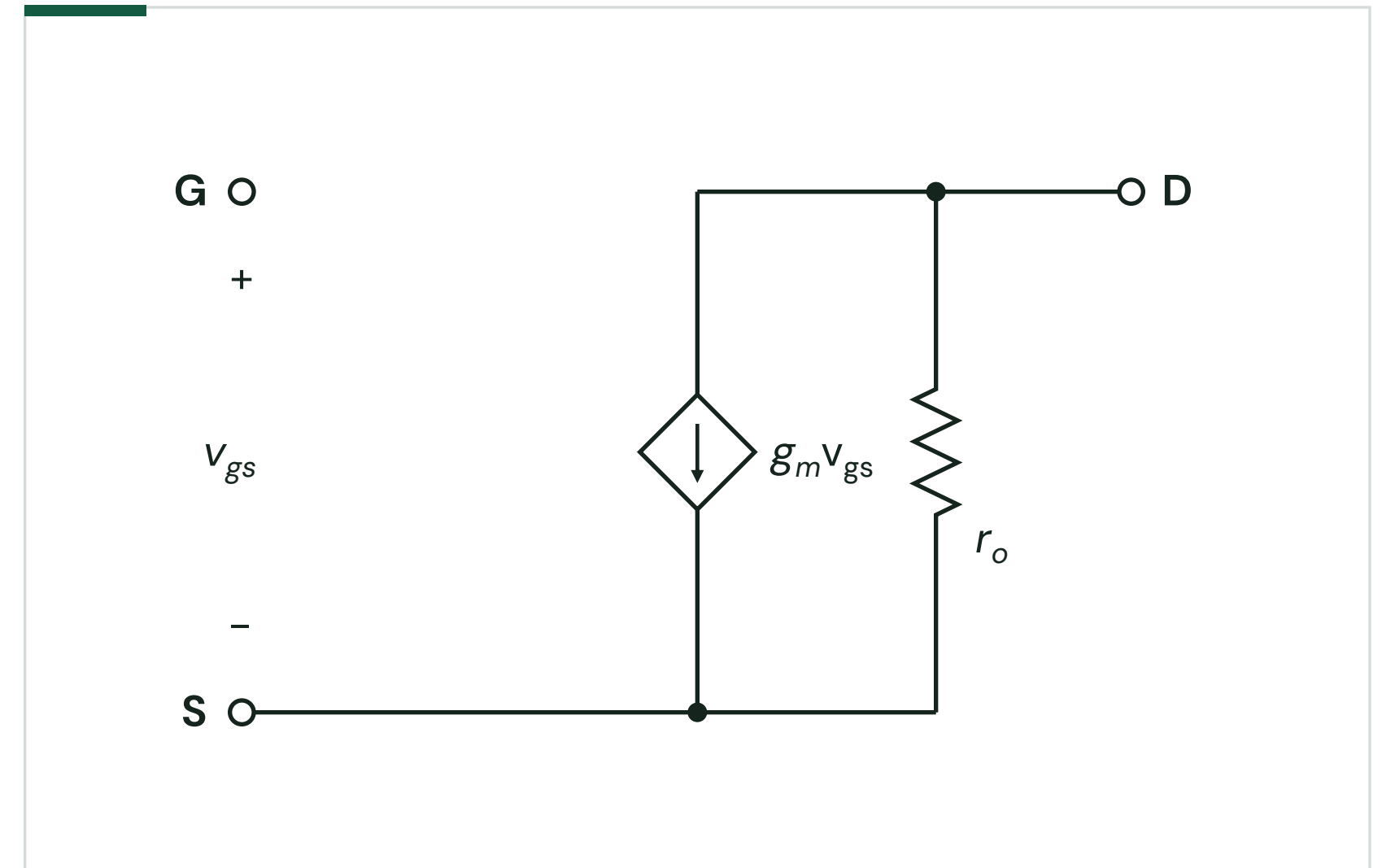
- 4 With a drain resistor R_D , the voltage gain is

$$A_v = \frac{v_{ds}}{v_{gs}} = -g_m R_D$$

SMALL-SIGNAL EQUIVALENT CIRCUITS

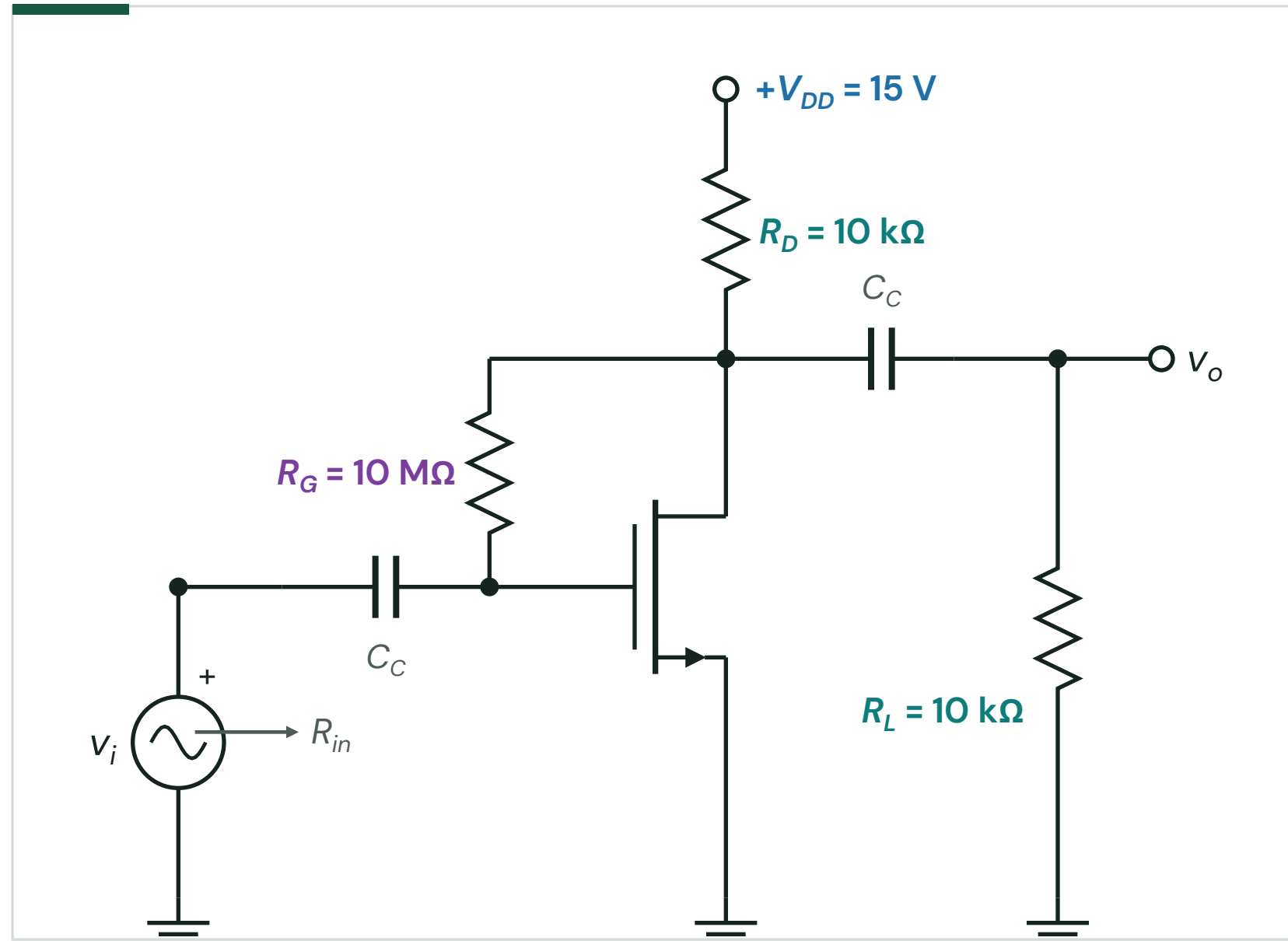


- The MOSFET as a voltage-controlled current source $g_m V_{gs}$; no current into the gate.



- Adding $r_o = V_A / I_D$ models the slight slope of the saturation curves (V_A : the Early voltage).

EXAMPLE 2: BIAS POINT (1 OF 2)



GIVEN

- $V_{DD} = 15 \text{ V}$
- $R_G = 10 \text{ M}\Omega$
- $R_D = R_L = 10 \text{ k}\Omega$
- $V_t = 1.5 \text{ V}$, $k'_n(\text{W/L}) = 0.25 \text{ mA/V}^2$, $V_A = 50 \text{ V}$

Find the gain $A_v = v_o/v_i$ and the input resistance R_{in} . First, the DC bias point.

1 Keep only the DC circuit

Coupling capacitors are open circuits at DC, so v_i , R_L and the output terminal drop out.

2 Relate V_{GS} to V_D

No gate current flows, so there is no drop across R_G : $V_G = V_D$. With the source grounded, $V_{GS} = V_{DS} \geq V_{GS} - V_t$; saturation is guaranteed.

$$V_{GS} = V_D = 15 - 10I_D$$

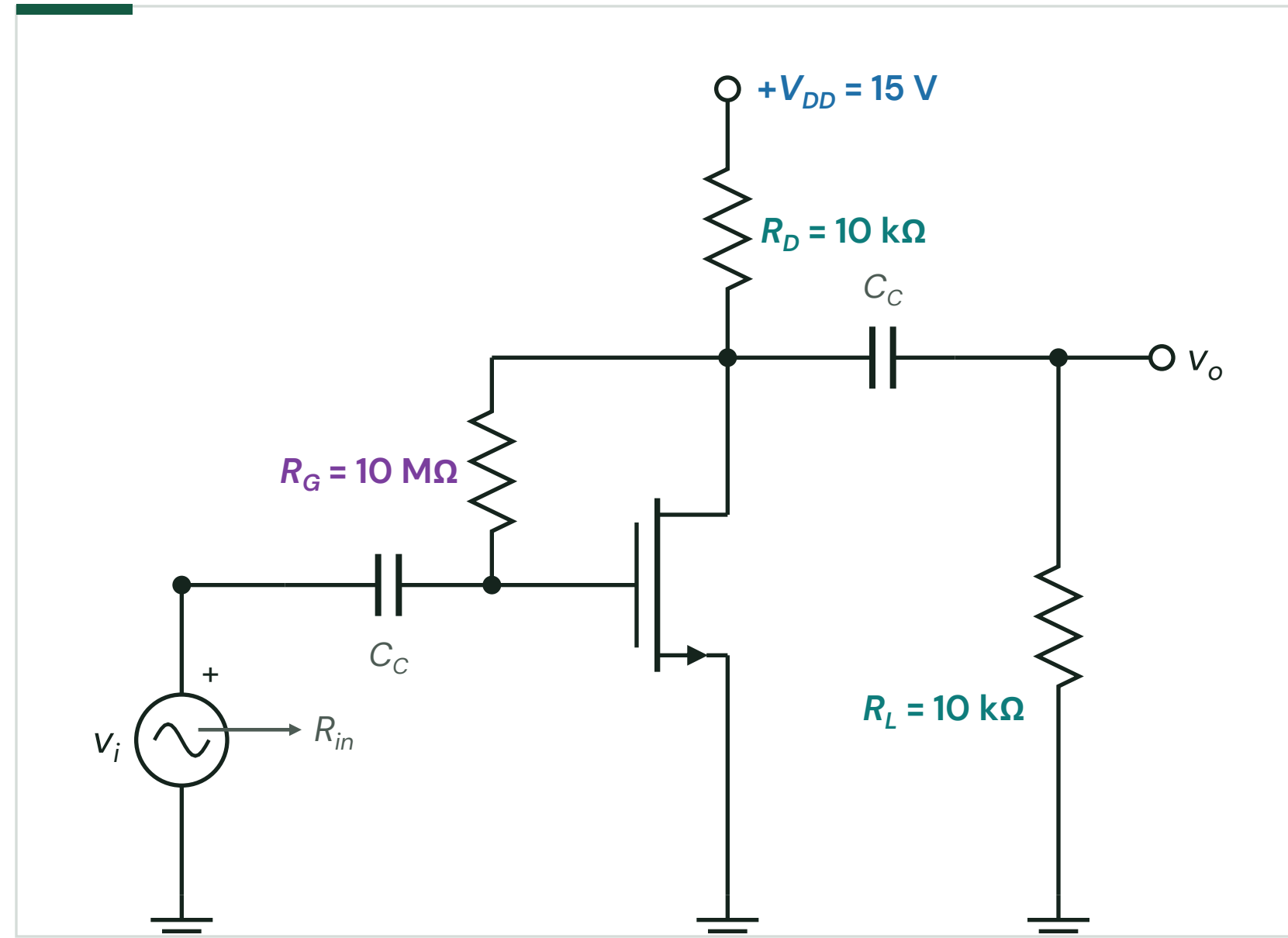
3 Solve for the bias current (I_D in mA)

$$I_D = \frac{1}{2} (0.25)(V_D - 1.5)^2$$

$$12.5I_D^2 - 34.75I_D + 22.78 = 0 \Rightarrow I_D = 1.06 \text{ or } 1.72 \text{ mA}$$

$I_D = 1.72 \text{ mA}$ gives $V_{GS} = -2.21 \text{ V} < V_t$; reject. So $I_D = 1.06 \text{ mA}$ and $V_{GS} = V_D = 4.41 \text{ V}$.

EXAMPLE 2: GAIN AND INPUT RESISTANCE (2 OF 2)



GIVEN

- $V_{DD} = 15 \text{ V}$
- $R_G = 10 \text{ M}\Omega$
- $R_D = R_L = 10 \text{ k}\Omega$
- $V_t = 1.5 \text{ V}$, $k'_n(W/L) = 0.25 \text{ mA/V}^2$, $V_A = 50 \text{ V}$

4 Small-signal parameters at the bias point

$$g_m = k'_n \frac{W}{L} (V_{GS} - V_t) = 0.25(4.41 - 1.5) = 0.728 \text{ mA/V}$$

$$r_o = \frac{V_A}{I_D} = \frac{50}{1.06} = 47.2 \text{ k}\Omega$$

5 Find the load seen by the drain

Capacitors are shorts for the signal. R_G is so large that its signal current is negligible.

$$R_D \parallel R_L \parallel r_o = 10 \parallel 10 \parallel 47.2 = 4.52 \text{ k}\Omega$$

6 Voltage gain

$$A_v = \frac{v_o}{v_i} \approx -g_m(R_D \parallel R_L \parallel r_o) = -0.728 \times 4.52 = -3.29 \text{ V/V}$$

7 Input resistance (Miller effect)

R_G joins input and output. The output swings by $A_v v_i$, so the current into R_G is $(1 - A_v)v_i/R_G$; the input sees R_G divided by $(1 - A_v)$.

$$R_{in} = \frac{R_G}{1 - A_v} = \frac{10}{1 + 3.29} = 2.33 \text{ M}\Omega$$

$$A_v \approx -3.3 \text{ V/V}$$

$$R_{in} = 2.33 \text{ M}\Omega$$

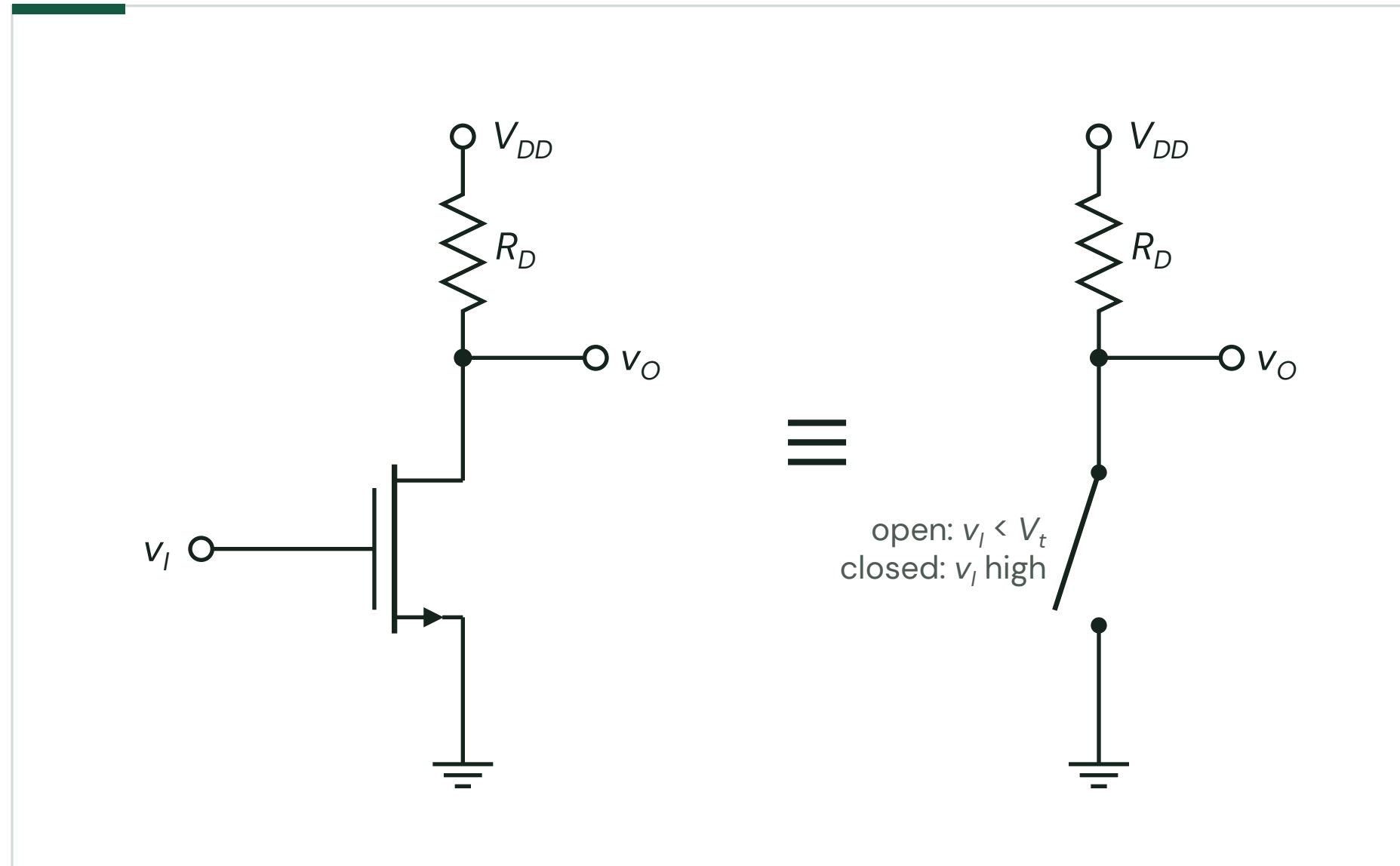
05

THE MOSFET AS A SWITCH

- Cutoff: an open switch
- Deep triode: a closed switch

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THE MOSFET AS A SWITCH



- $v_{GS} < V_t$: **cutoff**. The switch is **open** and $i_D = 0$.
- v_{GS} high and v_{DS} small: deep **triode**. The switch is **closed**, with a small resistance r_{DS} .
- Driving the gate between these two states is the basis of digital logic.

SUMMARY: THREE REGIONS OF OPERATION

REGION	CONDITION	DRAIN CURRENT
CUTOFF	$V_{GS} \leq V_t$	$i_D = 0$
TRIODE	$V_{GS} > V_t, V_{DS} < V_{GS} - V_t$	$i_D = k'_n \frac{W}{L} \left[(v_{GS} - V_t)v_{DS} - \frac{1}{2}v_{DS}^2 \right]$
SATURATION	$V_{GS} > V_t, V_{DS} \geq V_{GS} - V_t$	$i_D = \frac{1}{2}k'_n \frac{W}{L} (v_{GS} - V_t)^2$

- DC problems: **assume** a region, **solve**, then **check** the assumption.
- Small signals: $g_m = k'_n(W/L)(V_{GS} - V_t)$; a common-source stage has $A_v = -g_m R_D$.

THANK YOU

NEXT LECTURE

LECTURE 4: BIPOLAR JUNCTION TRANSISTORS (BJTS)

With thanks to **Dr. Sameh Osama Ezzat**, who prepared the original material for this lecture.

Figures: Sedra & Smith, Microelectronic Circuits (Oxford University Press); other diagrams from web sources via the original slides; circuits and the load-line plot redrawn for this lecture.